

The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease: meta-analysis of individual data from 27 randomised trials



Summary Background Statins reduce LDL cholesterol and prevent vascular events, but their net effects in people at low risk of vascular events remain uncertain. **Methods** This meta-analysis included individual participant data from 22 trials of statin versus control (n=134 537; mean LDL cholesterol difference 1.08 mmol/L; median follow-up 4.8 years) and five trials of more versus less statin (n=39 612; difference 0.51 mmol/L; 5.1 years). Major vascular events were major coronary events (ie, non-fatal myocardial infarction or coronary death), strokes, or coronary revascularisations. Participants were separated into five categories of baseline 5-year major vascular event risk on control therapy (no statin or low-intensity statin) (<5%, ≥5% to <10%, ≥10% to <20%, ≥20% to <30%, ≥30%); in each, the rate ratio (RR) per 1.0 mmol/L LDL cholesterol reduction was estimated. **Findings** Reduction of LDL cholesterol with a statin reduced the risk of major vascular events (RR 0.79, 95% CI 0.77–0.81, per 1.0 mmol/L reduction), largely irrespective of age, sex, baseline LDL cholesterol or previous vascular disease, and of vascular and all-cause mortality. The proportional reduction in major vascular events was at least as big in the two lowest risk categories as in the higher risk categories (RR per 1.0 mmol/L reduction from lowest to highest risk: 0.62 [99% CI 0.47–0.81], 0.69 [99% CI 0.60–0.79], 0.79 [99% CI 0.74–0.85], 0.81 [99% CI 0.77–0.86], and 0.79 [99% CI 0.74–0.84]; trend p=0.04), which reflected significant reductions in these two lowest risk categories in major coronary events (RR 0.57, 99% CI 0.36–0.89, p=0.0012, and 0.61, 99% CI 0.50–0.74, p<0.0001) and in coronary revascularisations (RR 0.52, 99% CI 0.35–0.75, and 0.63, 99% CI 0.51–0.79; both p<0.0001). For stroke, the reduction in risk in participants with 5-year risk of major vascular events lower than 10% (RR per 1.0 mmol/L LDL cholesterol reduction 0.76, 99% CI 0.61–0.95, p=0.0012) was also similar to that seen in higher risk categories (trend p=0.3). In participants without a history of vascular disease, statins reduced the risks of vascular (RR per 1.0 mmol/L LDL cholesterol reduction 0.85, 95% CI 0.77–0.95) and all-cause mortality (RR 0.91, 95% CI 0.85–0.97), and the proportional reductions were similar by baseline risk. There was no evidence that reduction of LDL cholesterol with a statin increased cancer incidence (RR per 1.0 mmol/L LDL cholesterol reduction 1.00, 95% CI 0.96–1.04), cancer mortality (RR 0.99, 95% CI 0.93–1.06), or other non-vascular mortality. **Interpretation** In individuals with 5-year risk of major vascular events lower than 10%, each 1 mmol/L reduction in LDL cholesterol produced an absolute reduction in major vascular events of about 11 per 1000 over 5 years. This benefit greatly exceeds any known hazards of statin therapy. Under present guidelines, such individuals would not typically be regarded as suitable

for LDL-lowering statin therapy. The present report suggests, therefore, that these guidelines might need to be reconsidered. Funding British Heart Foundation; UK Medical Research Council; Cancer Research UK; European Community Biomed Programme; Australian National Health and Medical Research Council; National Heart Foundation, Australia.